



CENTRAL UNIVERSITY OF SOUTH BIHAR

Department of Computer Science

Object Oriented Programming Methodology

Class Test 1

Duration: 1:00 hrs.

Maximum Marks: 10

Attempt all questions.

1. Write output of following program fragment:
(a) `a = (13, 3, 5, 6+3); a=?`
(b) `c= (a < b ? a : b); c=?` **(2 marks)**
2. Define operator overloading. Give an example. **(2 marks)**
3. Explain inline function. **(1 marks)**
4. Define a class student with data member as roll number and name. Generate an object and initialize its data member using default constructor, copy constructor and parameterized constructor. Finally, display all data members. **(5 marks)**



CENTRAL UNIVERSITY OF OUTH BIHAR

Department of Computer Science

Object Oriented Programming

Lab Test

Time: 1 hour

Total marks: 10

1. Create a class called Employee which contains protected attributes such as **emp_name**, **emp_id**, **emp_salary**, **emp_da** and **emp_hra**. **emp_da** and **emp_hra** are 20% and 10% of the **emp_salary**, respectively. Provide an appropriate method (you may use constructor) to take user input as **emp_name**, **emp_id** and **emp_salary** to initialize the attributes of an employee and display all attributes of corresponding employee.

OR

Write a program to overload operator '*' for complex number in C++ using Complex class. Complex class has two data members real and imaginary.



दक्षिण बिहार केन्द्रीय विश्वविद्यालय CENTRAL UNIVERSITY OF SOUTH BIHAR

Question Paper of End-Term Examination, May-2026

Name of Department: Computer Science

Programme: M.Sc. Computer Science

Session: 2025-2027

Semester: II

Course Title: Object Oriented Programming

Course Code: CSC82DC00704

Course Credit: 04

Max. Marks: 70

Duration: 3:00 Hrs

Enrolment No. (to be filed by student): CU5B2502312032

There are three sections (A, B, and C). Answer all questions as per instructions. Points against each question are mentioned in the parenthesis.

Section – A

Answer all the questions in not more than 50 words. Each question carries 1 Point.

(1 x 10 = 10)

1. Define virtual base class.
2. Write two differences between structure and union.
3. Explain pure virtual function.
4. What do you mean by instantiation in Template?
5. How will you dynamically allocate a memory in C++?
6. Explain polymorphism.
7. Write two differences between procedural concept and object-oriented concept.
8. Find value of z in the following program fragment:
 $x=10; y=20;$
 $z = (x < y ? x : y);$
9. What is static data member?
10. Explain function overloading.

Section – B

Answer ANY FIVE questions in not more than 300 words. Each question carries 6 Points.

(6 x 5 = 30)

1. Describe single inheritance, multiple inheritance, and multilevel inheritance by showing appropriate block diagrams.

2. What is a constructor? Explain different types of constructors with suitable examples.
3. Why does ambiguity arise in single inheritance? Mention mechanism to remove ambiguity in single inheritance with suitable example. [3+3 = 6]
4. What are the differences between a template and an overloaded function? Illustrate it by a suitable example.
5. Write a function template for finding the minimum value contained in an array of size N.
6. Write a program to overload == operator to check whether two circles are equal or not. (Note: Two circles are said to be equal if their radius is same and center has same coordinate)

Section – C

Answer ANY TWO questions in not more than 1000 words. Each question carries 15 Points. (15 x 2 = 30)

1. Create a base class shape. Use this class to store two double type values (x and y) that could be used to compute the area of figures. Derive two specific classes called triangle and rectangle from the base shape. Add to the base class a member function get_data() to compute and display the area of figures. Make display_area() as a virtual function and using these three classes design a program that will accept dimensions of a triangle or a rectangle interactively and display the area.
 Remember the two values given as input will be treated as length of two sides in the case of rectangle as base and height in case of triangle and used as follows:
 Area of triangle = $\frac{1}{2} * x * y$
 Area of rectangle = $x * y$
2. Write a program to create a class called a Person class. Each object of this class will represent a human being. Data members should include the person's name, year of birth, and year of death. The attributes should be initialized through default constructor, copy constructor and parameterised constructor. The class contains a public method named as show() to display the initialized attributes. Include destructor also in Person class.
3. What is an exception? Discuss mechanism to handle an exception in C++ in detail with a suitable C++ program/program fragment.